

Trajectory Classification with LSTM for Robotic Peg-in-Hole

Group 1 - LOUNAS Gana |
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UE5 Apprentissage

Master's Machine Learning
Project – 2025–2026





Project Objectives

- ▶ Build a trajectory-level classifier for a 7-DoF peg-in-hole task with **3 classes**.



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- ▶ Evaluation context: internal split (train/val/test), plus compatibility with an external test set.



Roadmap

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Dataset and Feature Blocks

- Files: `class1_trajectories.npy`, `class2_trajectories.npy`, `class3_trajectories.npy`

Block	Meaning	Size
peg_pose	peg position + quaternion	7
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Total		20



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- Reproducibility: fixed random seed (SEED=42) for split and initialization.

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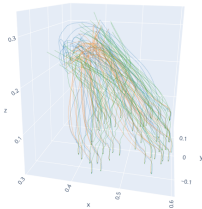
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- ▶ Numerical stability: $\sigma = 0$ replaced by small ϵ .

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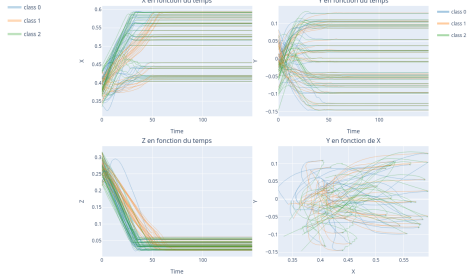


Trajectory Evidence Before Training

Trajectoires (position du peg) par classe



Trajectoires par composante



3D trajectories (x, y, z) by class.

Projections: $x(t), y(t), z(t), y(x)$.

Class-dependent temporal patterns are visible before training, which motivates a temporal classifier.



Why LSTM (Course Connection)

- Reminder: vanilla RNNs are sensitive to vanishing/exploding gradients through BPTT.

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- ▶ This architecture matches the project setting: long trajectories (150 steps) with dynamic class signatures.



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- ▶ Batch size: 32, training budget up to 40 epochs in main experiments.
- ▶ Stability control: gradient clipping $\|\nabla\|_{max} = 1.0$.



Controlled Experiment Plan

Run	Setting	Purpose
R1	hidden=64, 2 layers, 40 ep, no norm / dropout / clipping	raw baseline
R2	+ z-score + dropout(0.2)	normalization + regularization
R3	R2 + gradient clipping	training stability

- Fixed controls: seed, split protocol, optimizer family, class definition.



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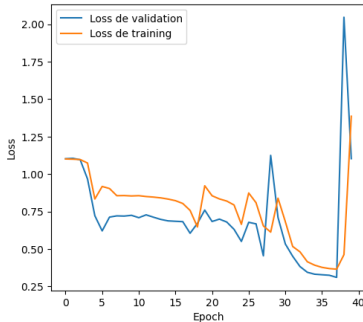
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- ▶ Fixed controls: seed, split protocol, optimizer family, class definition.
- ▶ Selection criterion: best generalization with stable learning curves.

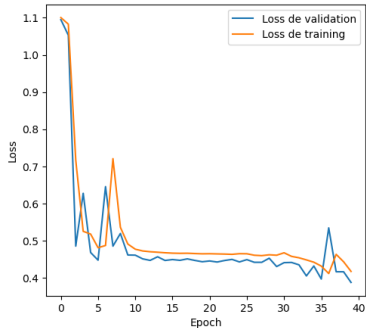


Practical Issues During Development

- **Reproducibility:** no fixed seed $\Rightarrow$ difficult run-to-run comparison.



No normalization: noisier and less predictable convergence.

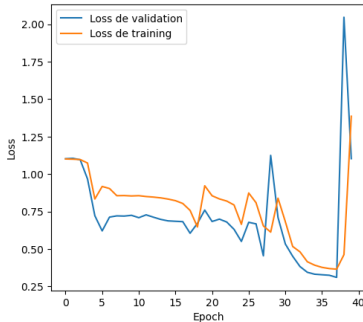


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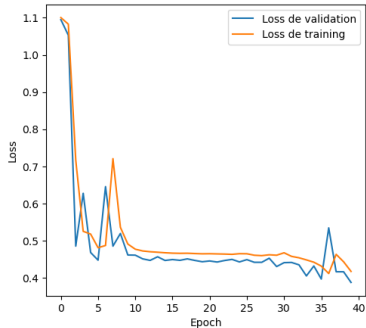


Practical Issues During Development

- ▶ **Reproducibility:** no fixed seed $\Rightarrow$ difficult run-to-run comparison.
- ▶ **Optimization sensitivity:** preprocessing and regularization strongly alter convergence quality.



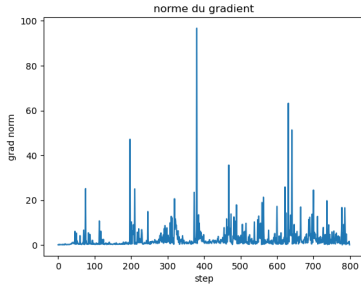
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Exploding Gradients and Mitigation



Gradient norm spikes observed during training.

Observed issue

Large gradient spikes destabilize the parameter's updates.

Fix used

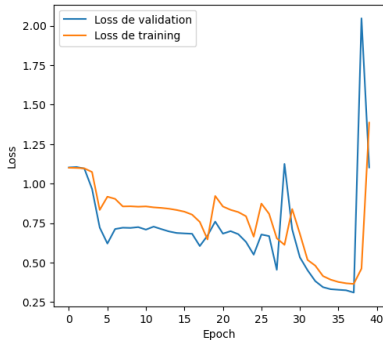
Gradient clipping in the training loop:
`clip_grad_norm(..., 1.0)`.

Effect

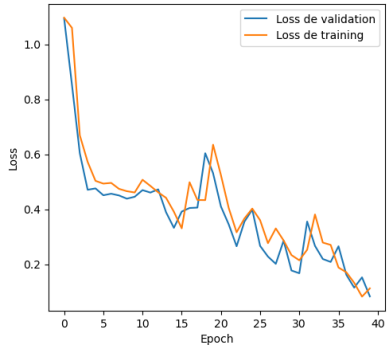
More stable updates and smoother final convergence.



Runs R1 and R2



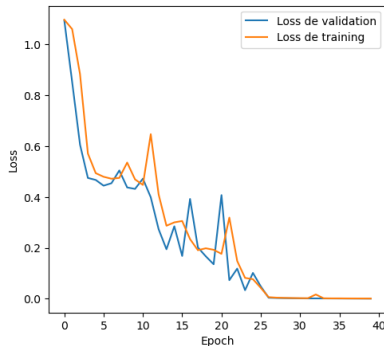
R1 baseline: loss diverges at the end —
no normalization, no dropout, no
clipping.



R2 (+z-score +dropout): **98.72%** test
accuracy (77/78).



Run R3 — Best Run



R3 (+dropout + gradient clipping): **100%** test accuracy (78/78).

- Gradient clipping eliminates late-training spikes and ensures stable convergence.
- This configuration is selected as the final model.



Quantitative Synthesis

Run	Test accuracy	Main effect
R1	diverges	no regularization or normalization
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- ▶ Most impactful additions in this pipeline: **normalization + dropout + clipping**.
- ▶ Internal split performance is excellent; external hidden test remains the final robustness check.



Feature-Ablation Protocol (Feature Notebook)

Subset	Indices	Epoch budgets
Peg pose + orientation	0:7	40, 80, 150
Peg linear+angular velocity	7:13	40, 80, 150
Hole pose + orientation	13:20	40, 150

- Goal: isolate which observation blocks truly carry discriminative information.



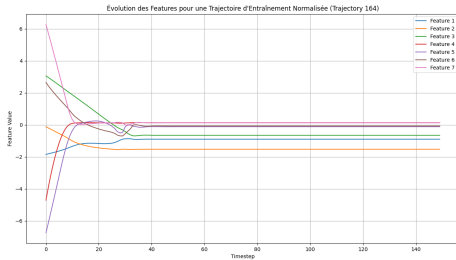
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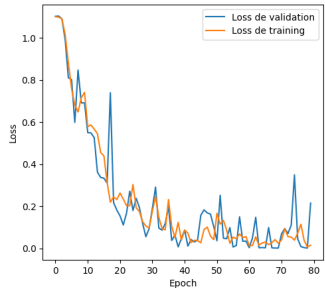
- ▶ Goal: isolate which observation blocks truly carry discriminative information.
- ▶ Same LSTM family and evaluation protocol; only feature subset and epoch budget vary.



Subset A: Peg Pose/Orientation Only (0:7)



Evolution of the 7 peg pose and orientation features.

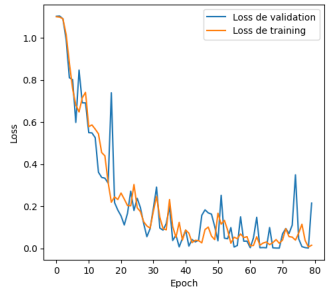
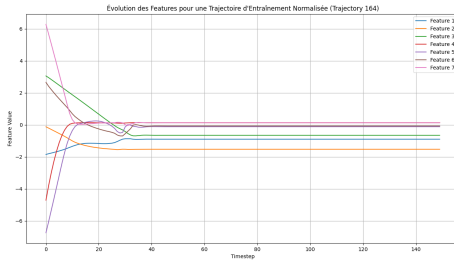


Example loss curve (80 epochs).

- Test performance: 98.72% (40 ep), 100% (80 ep), 100% (150 ep).



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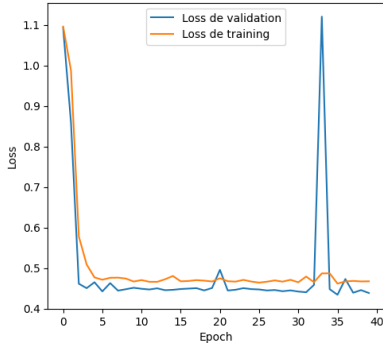
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- ▶ Interpretation: The peg's pose and orientation alone already contain most class-discriminative signal.

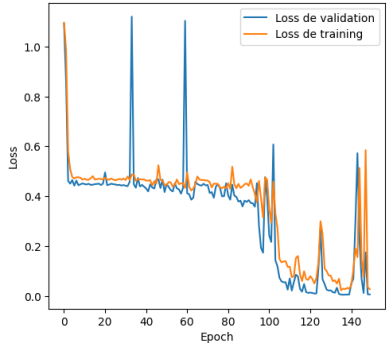
Feature Consequences Study



Subset B: Peg Velocities Only (7:13)



40 epochs: moderate, unstable plateau.

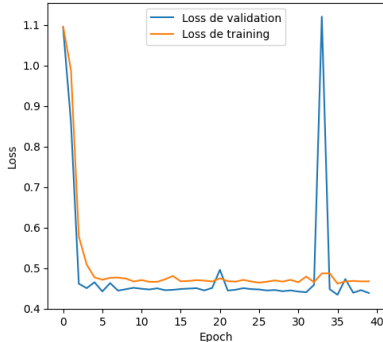


150 epochs: late transition to low loss.

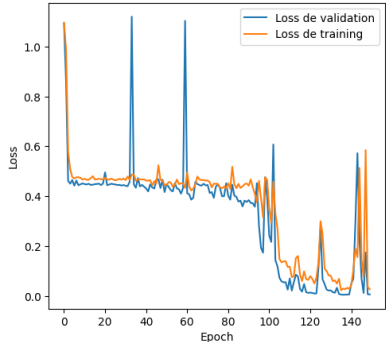
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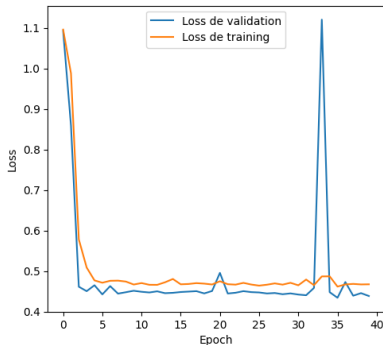


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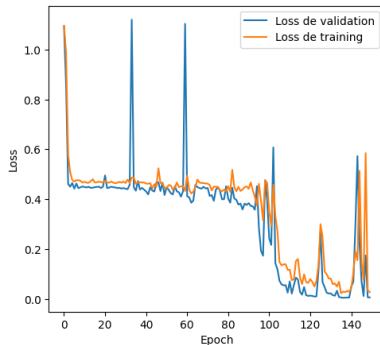
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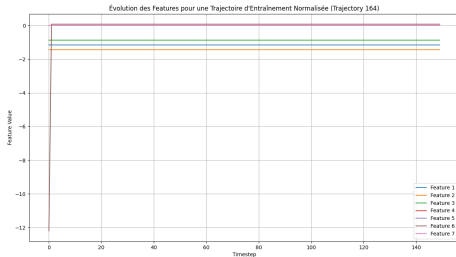


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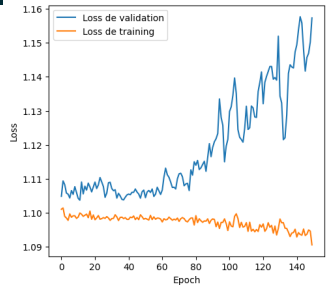
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- ▶ Velocity is informative, but optimization is harder and slower.



Subset C: Hole Pose/Orientation Only (13:20)



Hole-pose features remain constant over time.

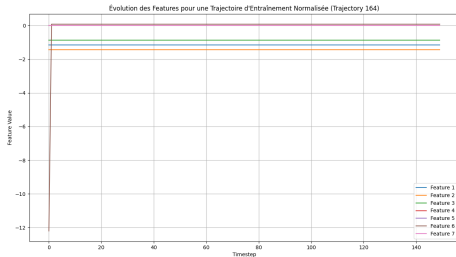


150 epochs: no meaningful learning.

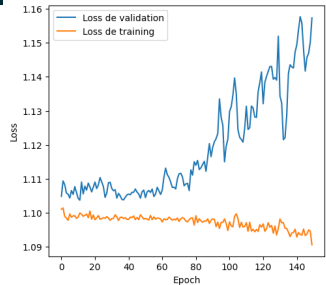
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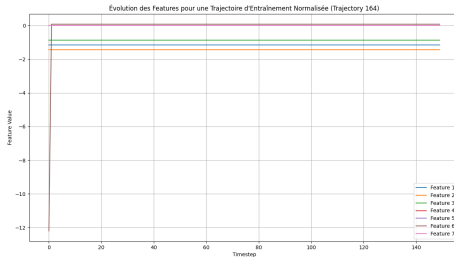


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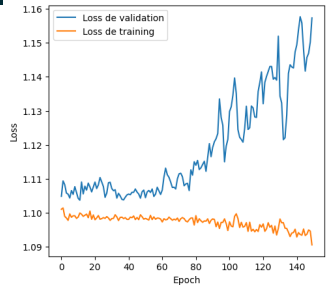
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- ▶ Interpretation: this subset is non-discriminative for trajectory classification.



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- ▶ Jury-level takeaway: good accuracy comes from both **signal selection** and **stable optimization**, not from architecture choice alone.



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- ▶ Jury-level takeaway: good accuracy comes from both **signal selection** and **stable optimization**, not from architecture choice alone.
- ▶ Final validation step remains the teacher-held hidden test set.



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- ▶ With 768 trajectories and 150 timesteps, over-capacity increases instability/overfitting risk.
- ▶ Two recurrent layers keep modeling depth while staying computationally light.



Backup: Model Size and Complexity

Component	Shape/setting	Parameters
LSTM layer 1	input 20, hidden 64	22,016
LSTM layer 2	input 64, hidden 64	33,280
Classifier head	Linear(64,3)	195
Total		55,491

- The model is compact ($\sim 55k$ params), which is appropriate for this dataset size.



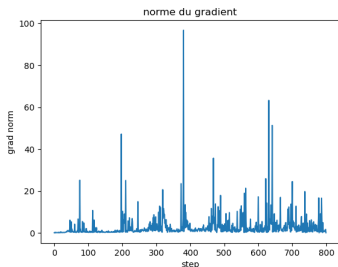
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- ▶ The model is compact ($\sim 55k$ params), which is appropriate for this dataset size.
- ▶ Capacity is sufficient to fit complex temporal patterns without requiring very large-scale tuning.



Backup: Why Gradient Clipping at 1.0?



Observed gradient norm spikes before stabilization.

Rule used

`clip_grad_norm(..., 1.0).`

Reasoning

- recurrent training showed large norm spikes
- only large gradients are rescaled

Observed effect

stable late-epoch behavior and no final divergence.



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- ▶ **Hole pose/orientation (13:20)**: near-constant in these trajectories, so weak discriminative power alone.



Backup: Feature Interpretation (Physical View)

- ▶ **Peg pose/orientation (0:7)**: directly tracks insertion geometry and trajectory signature.
- ▶ **Peg velocity (7:13)**: informative but noisier; requires longer optimization to fully separate classes.
- ▶ **Hole pose/orientation (13:20)**: near-constant in these trajectories, so weak discriminative power alone.
- ▶ Ablation behavior is therefore physically coherent with the task dynamics.



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- ▶ GRU/Transformer remain valid extension baselines for future comparison.